

Salient characteristics of an OVERHEAD CONVEYOR BLAST MACHINE for TLSC-E/A, Maintenance Activity Kaiserslautern (MAK)

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1. OVERVIEW

This requirement consists of purchase of one (01) each Overhead Conveyor Blast Machine.

The stationary overhead rail shot blasting system is designed for the mechanical surface treatment of steel components. This facility utilizes the turbine blasting process to work on items such as sheets, angle profiles, and welded steel constructions. It is suitable for use in non-ATEX environments and can process various part geometries.

Key Features of the System

- **Type:** Stationary Overhead Rail Shot Blasting System
- **Application Area:** Surface treatment of steel components
- **Quantity:** 1 unit
- **Dimensions:** High: max 6.000 mm Width: max 3.600 mm Length: max 7.000 mm

TECHNICAL REQUIREMENTS:

1. Workpiece Data / Application Range

- **Suitable for:** Steel parts of variable geometry
- **Types of Parts:** Sheets, angle profiles, welded assemblies
- **Minimum and Maximum Dimensions:** The effective diameter in the blasting area must be at least 900 mm.
- **Weight of Parts:** Less than 10 kg, fixation may be required.
- **Load Capacity:** Minimum load capacity per hook is 500 kg.

2. Blasting Chamber

- **Construction:** Stationary, highly wear-resistant
- **Interior:** Fully lined with replaceable wear protection material.
- **Safety:** Pneumatically locked safety door.
- **Opening Dimensions:** Minimum height of 1,500 mm and width of 1,100 mm.
- **Foundation:** A compensation frame must be provided for the jet system.

3. Workpiece Transport System (Overhead Rail)

- **Mobility:** Manual or fully automatic.
- **Oscillation and Rotation:** Integrated oscillation ± 150 mm, rotation 8 RPM adjustable).
- **Hook Retraction:** Automatic hook retraction before the blasting area.

4. Blasting Unit / Turbines

- **Number of Turbines:** At least 2 centrifugal turbines, each with ≥ 5.5 kW.
- **Throwing Speed:** ≥ 80 m/s.
- **Blast Media Throughput:** ≥ 110 kg/min per turbine.
- **Power Regulation:** Frequency converter for power regulation is mandatory.

5. Blast Media Recycling & Dosage

- **Feeding System:** Screw conveyor + bucket elevator, performance ≥ 500 kg/min.
- **Media Treatment:** Air classification / cascade separator.
- **Media Flow Regulation:** Automatically integrated.
- **Media Refill:** ≥ 40 liters integrated.

6. Dust Extraction / Filtration System

- **Filter Type:** Cartridge filter system.
- **Air Volume Flow:** $\geq 2,000$ m³/h.
- **Residual Dust Content:** ≤ 1 mg/m³.

7. Control / Operating Concept

- **Control System:** PLC-based (e.g., Siemens S7).
- **Operating Panel:** Touch or push-button display ≥ 7 inches.
- **Functions:** Error messages, operating hour meter, turbine current monitoring.
- **Remote Maintenance:** Via VPN connection (key switch activatable).
- **Power Supply:** 400 V / 50 Hz / 3 Ph N/PE.

8. Operating and Process Requirements

- **Workpiece Temperature:** $+5\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$.
- **Process Steps:** Texturing, derusting, cleaning.
- **Surface End Condition:** Free from loose scale, rust, or dirt particles.

1. COMMON SAFETY REQUIREMENTS FOR ALL EQUIPMENT

- The system must be brand new and CE compliant.
- EC Declaration of Conformity according to Machinery Directive is required.
- Operating, maintenance, and spare parts documentation must be in German.
- Circuit diagram according to DIN EN 60204-1.
- Warranty period of at least 12 months or 21,800 operating hours.
- Spare parts availability must be guaranteed for at least 10 years.
- Turbines and access doors secured with locking limit switches (DIN EN 1248)
- Compliance with DGUV Rule 100-500 Chapter 2.24 and DGUV Rule 113-001 regarding explosion-proof design of special filter systems for processing aluminum and magnesium
- DIN EN 1248 - Safety requirements for blasting machines
- Directive 2014/68/EU - Pressure Equipment Directive
- DIN EN 60204-1 - Electrical equipment of machines

2. COMMON REQUIREMENTS FOR ALL EQUIPMENT

- The warranty for the system, including all components and accessories, is 12 months from the date of delivery.
- 1 ea. Operation manuals, 1 ea. Maintenance manuals. All Documentation in English and German language.

3. TRANSPORT AND TRAINING REQUIREMENTS

- **Delivery:** Delivered to the point of use, including a compensation frame, unloading and ready-to-operate setup.
- **Commissioning:** By supplier.
- **Training:** Operator and maintenance training on-site in German, including certification.